

Effect of Different Nanoparticles on Tuning Electrical Properties of Polypropylene Nanocomposites

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ABSTRACT

Polypropylene (PP) has shown great potential as recyclable HVDC cable insulation material. This paper aims to investigate the effect of different nanoparticles on tuning the electrical properties of PP. Surface modified MgO, TiO₂, ZnO and Al₂O₃ nanoparticles with various contents were melt blended with PP. The microstructural morphology, dielectric properties, DC volume resistivity, space charge behavior and DC breakdown strength of the nanocomposites were investigated. It was found that the dielectric permittivity all increases with the increase of nanoparticle content, only the dielectric loss of TiO₂/PP nanocomposites increases. For MgO and TiO₂ nanocomposites, the DC volume resistivity increases with the increase of nanoparticle content and then decreases, while it continues to increase for Al₂O₃ nanocomposite. TiO₂/PP nanocomposite with 5 phr nano-TiO₂ shows lower volume resistivity than PP. Space charge suppression and DC electric breakdown strength show similar variation trend, both increase with the increase of nanoparticle content and then decrease. MgO and TiO₂ nanocomposites show the most obvious space charge suppression and TiO₂ nanocomposites exhibit the highest DC breakdown strength, which is 43% higher than that of pure PP. Considering the electrical properties investigated, the optimal content for MgO, TiO₂, ZnO and Al₂O₃ nanoparticles is about 3, 1, 1 and 1 phr, respectively. Among these four kinds of nanoparticles, MgO and TiO₂ nanoparticles are more capable than ZnO and Al₂O₃ nanoparticles to modify the electrical properties of PP and more potential to be used as recyclable HVDC cable insulation material.

Index Terms — Polypropylene, magnesium oxide, titanium oxide, zinc oxide, aluminum oxide, electrical properties, breakdown strength, volume resistivity, space charge.

1 INTRODUCTION

POLYMERIC insulation materials have been extensively used as high-voltage direct current (HVDC) cable insulation material for their low density, low dielectric constant and dielectric loss, excellent mechanical flexibility, ease of processing and most importantly, low cost [1-4]. The most widely used polymeric insulation materials in HVDC cables are low density polyethylene (LDPE) and cross-linked polyethylene (XLPE). XLPE is developed to compensate the poor mechanical properties of LDPE at high temperature by utilizing the crosslinked macromolecular networks in XLPE. But the crosslinking process also makes XLPE a thermoset material, which is difficult to be recycled at the end of its lifetime. Meanwhile, crosslinking byproducts such as phenol, cumyl alcohol, acetophenone and α -methyl styrene from the peroxides (e.g., dicumyl peroxide, DCP) would be introduced [5], which deteriorate the long-term performance of the insulation [6, 7]. Particularly, it has been shown that the

crosslinking byproducts could induce severe space charge accumulation, which makes XLPE not suitable for HVDC cable insulation application [8]. Recent advances in nanodielectrics have shown that introduction of nanoparticles can suppress the space charge accumulation and enhance the electrical properties of LDPE and XLPE, various kinds of nanoparticles such as MgO, SiO₂, TiO₂, Al₂O₃, ZnO, montmorillonite and zeolite have been introduced into LDPE and XLPE [9-15]. Although introducing nanoparticles can enhance the electrical properties of XLPE, there are limitations because XLPE is difficult to be recycled, which is not consistent with the concept of environmental protection and sustainable development. Therefore, it is necessary to develop potential recyclable HVDC cable insulation material to replace XLPE [16].

Polypropylene (PP), one of the most widely used commercial plastics, not only has much higher temperature tolerance and mechanical strength than LDPE and XLPE, but also has excellent electrical properties, such as high breakdown strength, high volume resistivity and low dielectric loss [17]. The most important thing is that polypropylene is a

thermoplastic material that can be easily recycled. Many researches have studied the fundamental properties of PP and other polypropylene-based thermoplastic materials and focused on the modification of mechanical properties of PP [18, 19]. Although modification of the electrical properties of LDPE, XLPE and epoxy by introducing nanoparticles has been widely studied, there is still limited knowledge about the effect of different nanoparticles on the properties of PP.

So it is important to study the effect of different nanoparticles on the electrical properties of PP to guide the development of potential recyclable HVDC cable insulation material. In this paper, four kinds of nanoparticles (MgO, TiO₂, ZnO and Al₂O₃) were introduced into PP with various mass fractions. The nanoparticles were surface modified with silane coupling agent to ensure good compatibility with PP matrix. The micro morphology, dielectric properties, DC volume resistivity, space charge characteristics and DC breakdown strength of the PP nanocomposites were studied and compared. This investigation would provide a reference for the design and development of potential recyclable HVDC cable insulation material.

2 EXPERIMENTAL PROCEDURE

2.1 MATERIALS

For the convenience of comparison, isotactic polypropylene (iPP) with a melt flow index of 3 g/10 min was used as the matrix material for its sample structure. Nanoparticles of MgO, TiO₂, ZnO and Al₂O₃ (Aladdin Industrial Inc., China) with an average diameter of 50 nm were used as the nano-fillers. The silane coupling agent γ -methacryloxypropyltrimethoxysilane (trade name KH570) was provided by Sinopharm Chemical Reagent Co. Ltd, China. Toluene of analytical pure grade was also obtained from Sinopharm Chemical Reagent Co. Ltd, China. Table 1 lists the properties of the nanoparticles.

Table 1. Properties of the nanoparticles.

Filler type	Measured specific surface area (m ² /g)	Relative dielectric constant	Resistivity (Ω -m)
MgO	21	9.7	$>10^{12}$
TiO ₂	73	48	$>10^{11}$
ZnO	48	1.7-2.5	10^8 - 10^{10}
Al ₂ O ₃	25	9.4	10^{12} - 10^{14}

2.2 SURFACE MODIFICATION OF NANOPARTICLES

Surface modification of the nanoparticles with silane coupling agent was carried out as follows. The nanoparticles were vacuum dried at 100 °C for 12 hours before modification and then dispersed in toluene by sonication for 30 minutes. Afterwards, KH570/toluene mixture was slowly added into the suspension of nanoparticles/toluene and the suspension was refluxed with magnetic stirrer at 120 °C for 6 hours. Then, the nanoparticles were obtained by centrifugation and washed with fresh toluene for 3 times to remove the excessive silane coupling agent. Finally, the surface modified nanoparticles were dried in vacuum oven at 60 °C for 12 hours.

2.3 PREPARATION OF NANOCOMPOSITES

PP pellets and the surface modified nanoparticles were dried in vacuum oven at 60 °C for 12 hours prior to being blended in a mixer (Haake PolyLab QC, Germany) at 200 °C for 12 minutes. The rotation speed of the mixing rotor was 60 rpm. The masterbatch with 9 phr surface modified nanoparticles was firstly obtained and then diluted into the nanocomposites with 1, 3, and 5 phr nanoparticles (here 1 phr means 1 g nanoparticles in 100 g PP matrix). Pure PP without any nanoparticles was also melt blended in the mixer for comparison. Film samples with different thickness for characterization were prepared by compression molding at 200 °C under a pressure of about 15 MPa for 8 minutes and then cooled to room temperature with circulating water under the same pressure. All obtained samples were stored in the vacuum drier to exclude the influence of moisture prior to measurements.

2.4 CHARACTERIZATION

Chemical structure of the nanoparticles with and without surface modification was analyzed by X-ray photoelectron spectroscopy (XPS, Thermo Fisher Escalab 250XI, USA) with an Al K α source. The binding energy range was 0-1350 eV.

Fourier-transform infrared (FT-IR) spectrometer (Thermo Fisher Nicolet iS10, USA) was used to characterize the surface modification of the nanoparticles between 650 and 4000 cm⁻¹ with the resolution of 2 cm⁻¹.

Thermogravimetric analysis (TGA) of the nanoparticles with and without surface modification was conducted on a TA Q500 instrument (TA Instruments, USA) at a heating rate of 10 °C/min from room temperature to 800 °C under nitrogen atmosphere.

The morphology and microstructure of the nanocomposites were investigated by field emission scanning electron microscope (FE-SEM, Hitachi 8010, Japan). Samples for FE-SEM observation were broken in liquid nitrogen and sputtered with platinum on the fractured surface to avoid charge accumulation while observing.

DC volume resistivity was measured with a digital high resolution electrometer (Keithley 2635B, USA) equipped with standard three-electrode system. The measurements were performed on the samples with the thickness of around 120 μ m under 60 kV/mm DC electric field. Before the DC voltage was applied, the samples were electrically short-circuited for 5 minutes and then the leakage current was recorded immediately after the voltage was applied. The DC volume resistivity was calculated according to the leakage current at 10 minutes after the DC voltage was applied and five specimens were measured for repeat measurements.

Dielectric permittivity and loss tangent of the nanocomposites were measured with an Alpha-A high performance frequency analyzer (Novolcontrol GmbH Concept 40, Germany). Samples were about 250 μ m in thickness and sputter-coated with gold on both sides to serve as electrodes. The tests were carried out in the frequency range of 10⁻¹-10⁶ Hz at room temperature.

DC electric breakdown tests were performed on a dielectric strength tester (HT-50, Guilin Electrical Equipment Scientific Research Institute, China) using film samples with thickness of around 120 μm at room temperature. The samples were sandwiched between two 10 mm ball electrodes and immersed in silicone oil to prevent surface flashover. An increasing DC voltage of 1 kV/s was applied until the samples were breakdown. For each sample, 20 points were tested and the test data were processed using two-parameter Weibull statistical distribution method according to IEEE Std 930-2004.

Space charge distribution within the samples under DC electric field was measured using the pulsed electro-acoustic (PEA) method at room temperature. Samples for PEA measurements were about 250 μm in thickness and the DC electric field of 60 kV/mm was applied for 30 minutes. The reproducibility of the experimental profiles was confirmed by repeating the experiments at least three times for each sample.

3 RESULTS AND DISCUSSION

3.1 CHARACTERIZATION OF NANOPARTICLE SURFACE MODIFICATION

The physical and chemical properties of nanoparticle surface would influence the microstructure and properties of the nanocomposites, especially the dispersion of the nanoparticles [20]. The nanoparticles used in this paper were surface modified with silane coupling agent KH570 to increase the compatibility with PP matrix. XPS was employed to investigate the results of surface modification and the detailed surface chemical structure of the surface modified nanoparticles. As the XPS spectra shown in Figure 1, the appearance of new Si_{2p} peak at about 102 eV and the increased intensity of C_{1s} peak at about 295 eV in the surface modified nanoparticles confirm that the silane coupling agent has been successfully grafted onto the nanoparticle surface [21]. In order to obtain more clear insight, the Si_{2p} peak between 96 eV and 110 eV in the XPS spectra is enlarged in Figure 1.

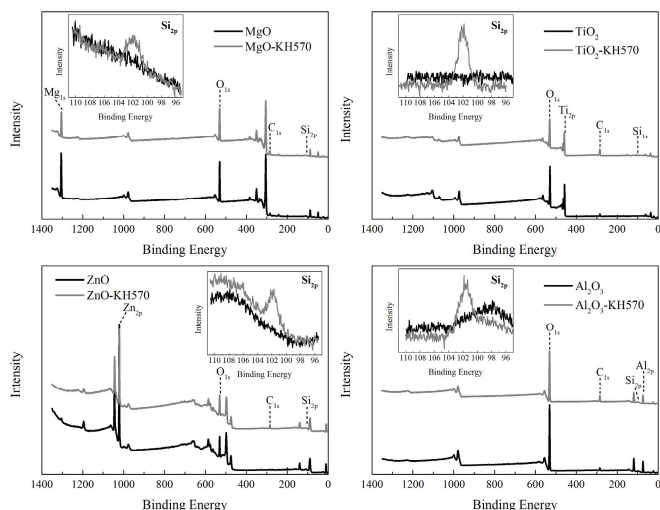


Figure 1. XPS spectra of the different nanoparticles with and without surface modification.

FT-IR and TGA curves shown in Figures 2 and 3 provide complementary evidence that the silane coupling agent was successfully grafted onto the nanoparticle surface. In the FT-IR spectra of surface modified nanoparticles, new absorption peaks appear at 2920, 2850 and 1720 cm^{-1} , corresponding to the $-\text{CH}_2-$, $-\text{CH}_3$ and $\text{C}=\text{O}$ groups, which are associated with the groups in the KH570. The surface modified nanoparticles all show new weight loss peak at higher temperature compared to the as-received nanoparticles, which should be attributed to the loss of organic groups grafted onto the nanoparticle surface. So, it can be concluded that the nanoparticle surface modification is successful but the goodness of coverage is hard to study.

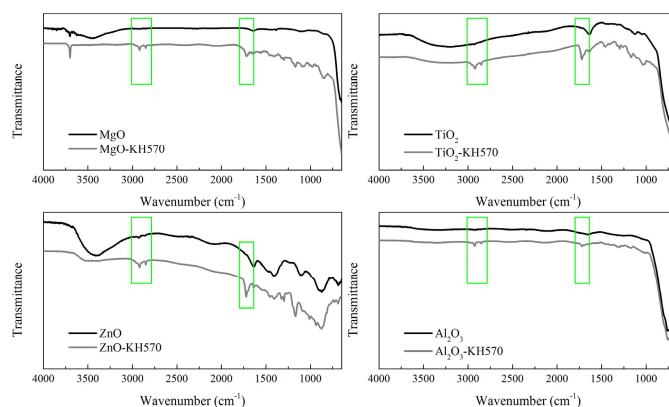


Figure 2. FT-IR spectra of the different nanoparticles with and without surface modification.

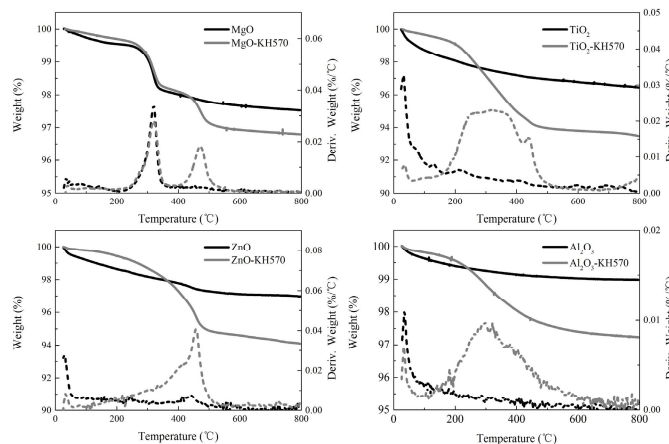


Figure 3. TGA curves of the different nanoparticles with and without surface modification. (solid lines for Weight and dash lines for Deriv. Weight).

3.2 MORPHOLOGY OF THE NANOCOMPOSITES

The freeze-fractured cross sections of the nanocomposites were observed by SEM to characterize the dispersion of the different nanoparticles in PP matrix. Figure 4 shows the SEM images of the PP nanocomposites with 3 phr surface modified nanoparticles. The different surface modified nanoparticles are all well distributed in PP matrix with the dimension of less than 250 nm without any large aggregates. It is believed that surface modification could change the chemical and physical properties of the nanoparticle surface and the surface modified nanoparticles would exhibit excellent compatibility with

polymer matrix [20]. The average diameter of the nanoparticles varies in the nanocomposites with different nanoparticles and the average diameter of the nanoparticles in Figure 4 is about 230 nm, 120 nm, 210 nm and 140 nm in the MgO, TiO₂, ZnO and Al₂O₃ nanocomposites, respectively.

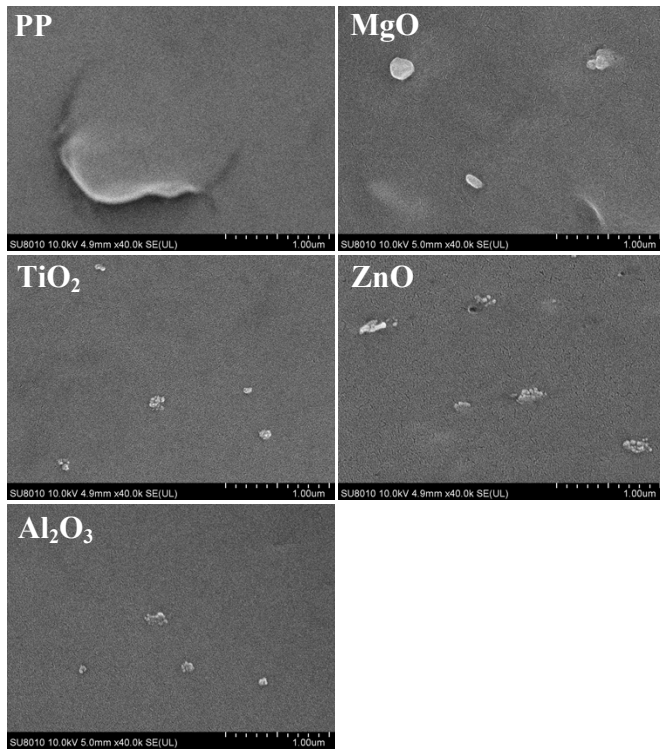


Figure 4. SEM images of PP nanocomposites with 3 phr different nanoparticles.

3.3 DIELECTRIC PROPERTIES

Figure 5 shows the frequency dependent dielectric permittivity and loss tangent of the PP nanocomposites with different nanoparticles and various contents. For the nanocomposites with certain kind of nanoparticle, the dielectric constant increases with the increase of nanoparticle

content. It is believed in literature that there are two main reasons for the changes of dielectric constant in nanocomposites, one is that the interaction between nanoparticles and polymer matrix would restrict the movement of the polymer molecular chains and reduce the polarization of polymer matrix [2, 22]. The other is that the interfacial polarization may exist at the interfaces between nanoparticles and polymer matrix, which would increase the dielectric constant [10]. The dielectric constant of the nanocomposites is influenced by the above two factors, which cause two opposite results and changes of the dielectric constant depend on which factor is dominant. In this investigation, maybe interfacial polarization is the dominant factor that increases the dielectric constant with the increase of nanoparticle content. Also, the nanoparticles used in this paper all have larger dielectric constant than the PP matrix and this could also result in the increased dielectric constant. But there are also some results that inclusion of nanoparticles can decrease the dielectric constant as the result of polymer chain movement restriction by nanoparticles [2]. Effect of nanoparticles on the dielectric constant is very complicated and there is still no consensus. Previous study on PP/nanoclay based nanocomposites has revealed that the dielectric constant of the nanocomposites is increased [23].

Although the dielectric constant increases with the increase of nanoparticle content in all MgO, TiO₂, ZnO and Al₂O₃ nanocomposites, the characteristic varies with different nanoparticles. Dielectric constant of MgO and Al₂O₃ nanocomposites only slightly increases, which should be attributed to the low dielectric constant of these two kinds of nanoparticles. Also, the MgO and Al₂O₃ nanoparticles are almost spherically distributed in PP with large surface area that hinders the molecular chain movements and thus limits the increase of dielectric constant. For ZnO nanocomposites, the nanoparticles are more likely distributed in ellipsoidal or fusiform, which have less interfaces than the spherical nanoparticles and the dielectric constant increase in ZnO nanocomposites is obvious. Among these four kinds in

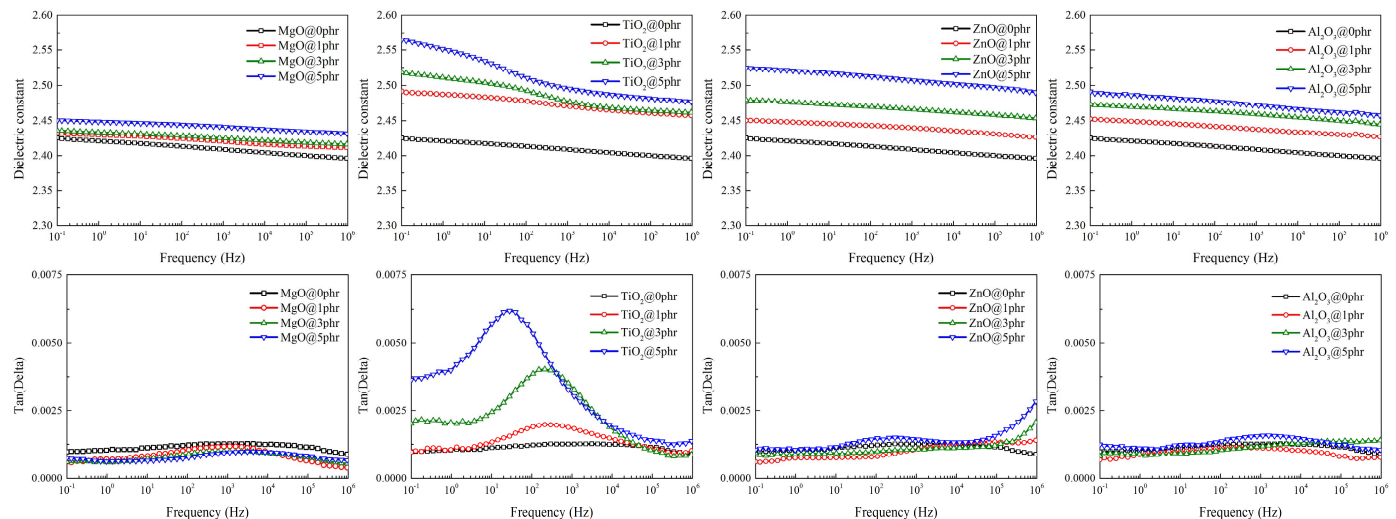


Figure 5. Dielectric spectra of PP nanocomposites with different nanoparticles.

nanocomposites, changes of dielectric constant of TiO_2 nanocomposites are the most special because with the increase of TiO_2 nanoparticle content, dielectric constant at lower frequency increases obviously. This can be explained that TiO_2 nanoparticles have the highest dielectric constant and the interfacial polarization plays an important role in TiO_2 nanocomposites, especially at lower frequency.

Figure 5 also shows that the frequency dependent feature of dielectric constant in MgO , ZnO and Al_2O_3 nanocomposites is weak because of the non-polar nature of PP. However, dielectric constant of TiO_2 nanocomposites increases at lower frequency, especially at higher TiO_2 nanoparticle content. As explained above, interfacial polarization in TiO_2 nanocomposites is more obvious, which has frequency dependent feature. Except for TiO_2 nanocomposites, the loss tangent of MgO , ZnO and Al_2O_3 nanocomposites all increase with the increase of nanoparticle content but are all lower than that of pure PP. While for TiO_2 nanocomposites, the loss tangent is always larger than pure PP and increases with the increase of nanoparticle content and shows a peak at lower frequency, which is attributed to the interfacial polarization discussed above.

3.4 DC VOLUME RESISTIVITY

DC volume resistivity is a key parameter for insulation materials, which determines the dielectric loss under DC electric field. Figure 6 gives the DC volume resistivity of the PP nanocomposites with different nanoparticles and various contents. It can be seen that introduction of different kinds of nanoparticles could all increase the DC volume resistivity of PP but the degree of the increase differs. For MgO and ZnO nanocomposites, the DC volume resistivity firstly increases with the increase of nanoparticle content and reaches its maximum value at 3 phr and then decreases. The DC volume resistivity of MgO and ZnO nanocomposites with 3 phr nanoparticles is about 3 and 3.5 times larger than that of pure PP, respectively. While DC volume resistivity of Al_2O_3 nanocomposites continues to increase with the increase of nanoparticle content and the maximum value is about 3.6 times larger than pure PP. Similar results of increased DC

volume resistivity have been reported in various kinds of polymer nanocomposites [9, 12, 13, 17, 24]. These previous researches have pointed out that inclusion of MgO , ZnO and Al_2O_3 nanoparticles would introduce numerous deep traps at the interfaces between nanoparticles and polymer matrix and the deep traps could capture the charge carriers and reduce the charge carrier mobility, thus increase the DC volume resistivity [25, 26]. But for TiO_2 nanocomposites, only the DC volume resistivity of the nanocomposites with 1 phr nanoparticles slightly increases and then decreases with the increase of nanoparticle content.

3.5 SPACE CHARGE BEHAVIOR

For HVDC insulation materials, space charge accumulation is an extremely important parameter that affects the long-term performance and serving life of the insulation and the apparatus. So space charge accumulation characteristics of the PP nanocomposites with various nanoparticles are studied. Figure 7 shows the dynamic space charge accumulation of different samples under 60 kV/mm DC electric field for 30 minutes at room temperature. As clearly shown in Figure 7, homocharges are injected at the cathode in pure PP immediately after the DC voltage is applied. Also the charge injection depth in pure PP increases with the increase of polarization time. These injected homocharges would cause local electric field distortion, especially the electric field near the anode would be increased, which is not favorable for DC insulation applications.

While with the inclusion of surface modified MgO , TiO_2 , ZnO and Al_2O_3 nanoparticles, space charge accumulation in the nanocomposites all can be suppressed but the inhibition effect depends on the type of the nanoparticles. MgO , TiO_2 and Al_2O_3 nanocomposites show the most desirable space charge suppression with lower nanoparticle content of under 3 phr and this is consistent with previous results in MgO/LDPE , MgO/XLPE , TiO_2/LDPE and $\text{Al}_2\text{O}_3/\text{LDPE}$ nanocomposites [12, 27, 28]. But for the doping of ZnO nanoparticles, space charge suppression is very limited. Homocharge injection at the cathode still appears and only the injection depth is reduced with 1 and 3 phr ZnO nanoparticles introduced. While in the previous study of ZnO/LDPE nanocomposite, significant space charge suppression was observed in the nanocomposite with 7 wt% KH550 surface modified ZnO nanoparticles with the diameter of 30 nm [25]. Such difference is suspected to be caused by the different shape, diameter and content of the nanoparticles as well as the different surface modification methods which result in different adhesion properties of the nanoparticles. For the nanocomposites with 5 phr nanoparticles, all four kinds of samples exhibit deteriorated space charge accumulation, the nanocomposite with 5 phr ZnO nanoparticles is even worse than pure PP. This should be attributed to the appearance of nanoparticle agglomeration which is caused by large amount of nanoparticles introduced [24].

As the field distortion caused by space charge accumulation is a great concern, the electric field distribution within the samples is calculated according to

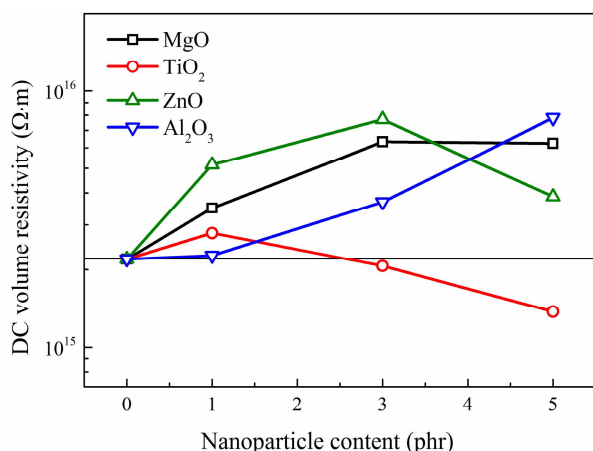


Figure 6. DC volume resistivity of PP nanocomposites with different nanoparticles.

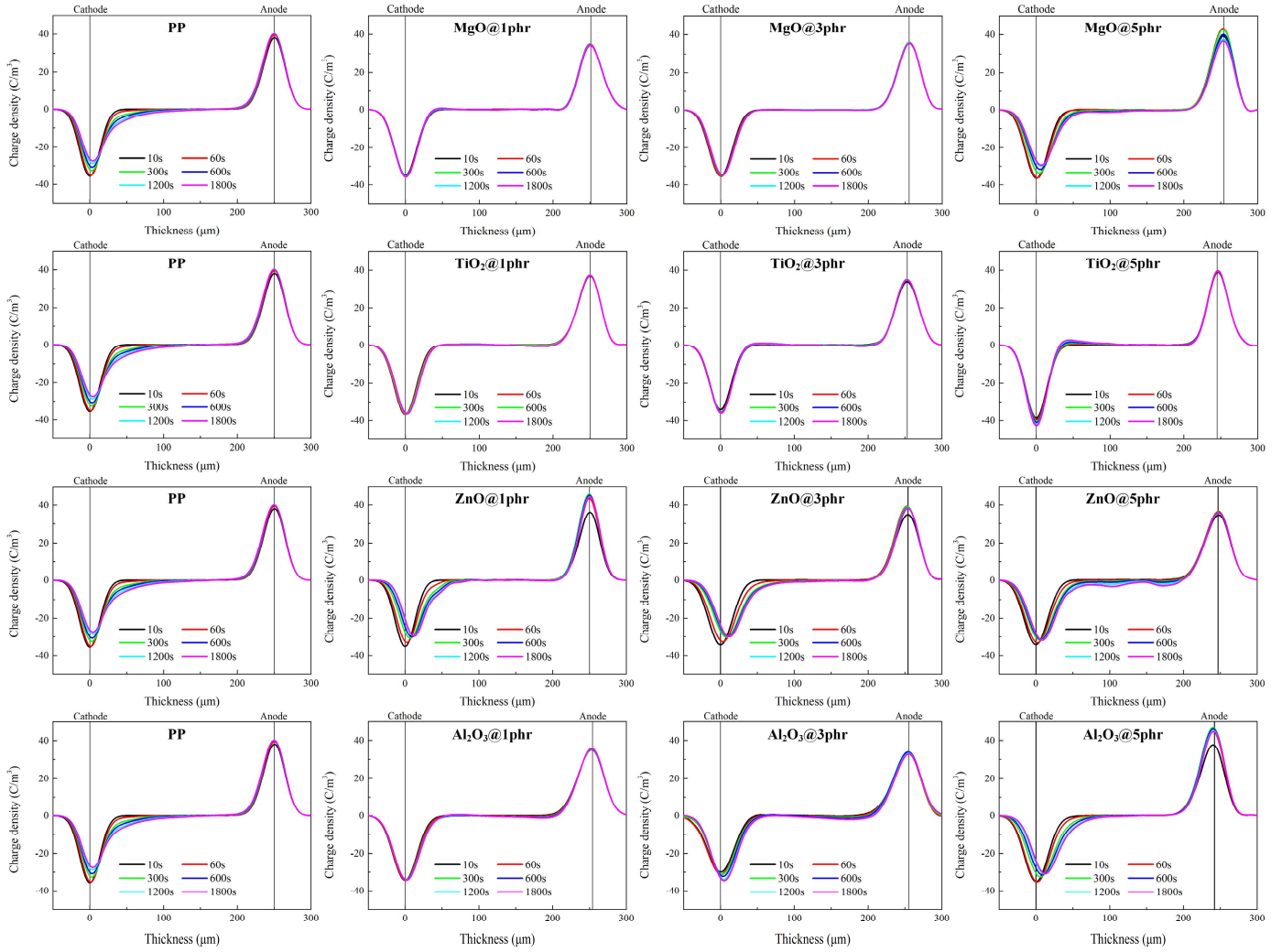


Figure 7. Space charge distribution in PP nanocomposites with different nanoparticles under 60 kV/mm electric field.

$$E(x) = \frac{1}{\epsilon_r \epsilon_0} \int_0^d \rho(x) dx \quad (1)$$

where $\rho(x)$ is the measured charge density distribution, ϵ_r is the relative dielectric permittivity of the sample, ϵ_0 is the vacuum dielectric permittivity and d is the distance between the two electrodes. Figure 8 shows the electric field distribution in different samples and Figure 9 shows the electric field distortion factor of the samples. The distortion factor is defined as

$$\Delta E = \left| \frac{E_{max} - E_{av}}{E_{av}} \right| \times 100\% \quad (2)$$

where E_{max} is the highest electric field within the samples at 30 minutes after the DC voltage is applied, E_{av} is the applied average electric field, which is 60 kV/mm in this paper. The distortion factor of pure PP is about 16%. While except for ZnO/PP nanocomposites with 5 phr nanoparticles, all other nanocomposites show lower distortion factor than pure PP. MgO and TiO₂ nanocomposites show the lowest distortion factor which are all lower than 7% and this is very favorable for industrial applications. Because high local electric field distortion may lead to partial discharge within the insulation materials and initial electrical treeing propagation.

3.6 DC BREAKDOWN STRENGTH

DC breakdown strength is the most important electrical parameter for dielectric materials in engineering. The Weibull plots of DC electric breakdown strength of pure PP and the nanocomposites with different nanoparticles are shown in Figure 10a-10d. The characteristic breakdown strength E_0 at 63.2% cumulative breakdown probability of the samples is shown in Figure 10e. The characteristic DC breakdown strength of pure PP is 389 kV/mm, which is much higher than that of LDPE and XLPE [16]. Compared with pure PP, introduction of various nanoparticles all increases the DC breakdown strength of PP. Changes of DC breakdown strength with the nanoparticle content all show the same characteristics that increases at first and then decreases with the increase of nanoparticle content, and most of the nanocomposites exhibit higher DC breakdown strength compared with pure PP. The optimal content of increasing DC breakdown strength of PP by introducing MgO, TiO₂, ZnO and Al₂O₃ nanoparticles is 3, 1, 1 and 1 phr, respectively, which is consistent with the optimal content of space charge suppression.

The effect of different nanoparticles on improving the DC breakdown strength and the optimal nanoparticle content are

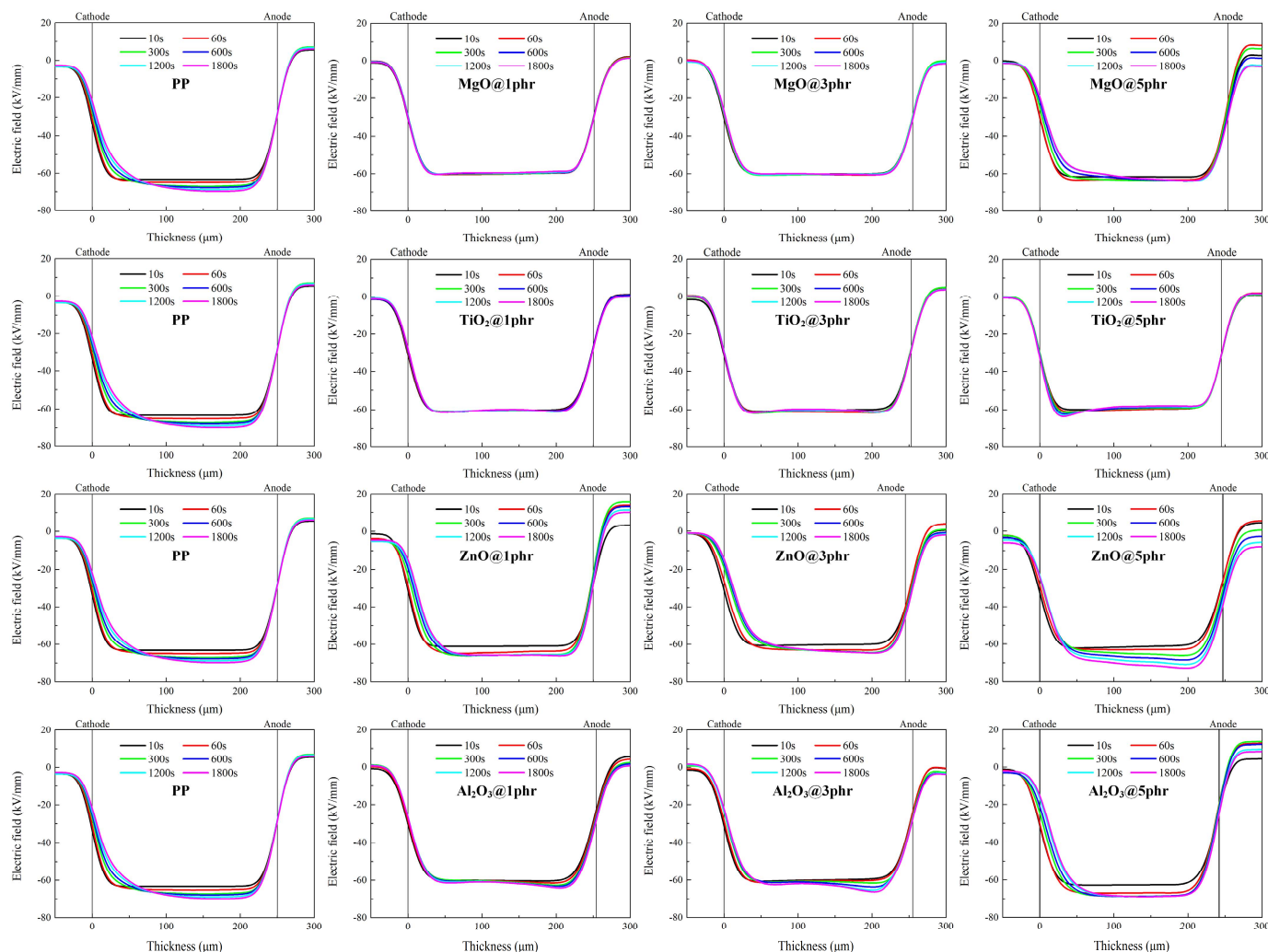


Figure 8. Electric field distribution in PP nanocomposites with different nanoparticles under 60 kV/mm electric field.

influenced by various parameters, such as the permittivity, conductivity, shape and dispersion of the nanoparticles as well as the compatibility of the nanoparticles and polymer matrix [29]. Among these four kinds of nanoparticles used in this study, TiO_2/PP nanocomposites with 1 phr nanoparticles show the highest DC breakdown strength of 557 kV/mm, which is 43% higher than that of pure PP. The highest DC breakdown strength of MgO, ZnO and Al_2O_3 nanocomposites is 503, 420 and 458 kV/mm, with the enhancement of 29%, 8% and 18% compared with pure PP.

Previous studies have shown that short-term electric breakdown is mainly an electronic process, which mainly depends on two factors, density of the free charge carriers and the energy of the charge carriers could obtain. The charge carriers would come from the charge injection at the electrodes or impurity ionization. The energy of the charge carriers could be accelerated in the DC electric field. If the charge carriers are accelerated to enough high energy, they would strike and break the polymer molecular chains and cause initial partial discharge and then lead to the erosion of the polymer material and final breakdown [30]. In this view, factors that decrease the free charge carrier density and charge

carrier mobility can contribute to the increase of electric breakdown strength. So the increase of DC electric breakdown strength by introducing nanoparticles can be explained in following aspects.

As explained in section 3.4, deep traps may be introduced at the interfaces between nanoparticles and polymer matrix, which decrease the mobility of charge carriers. The nanoparticles can also act as scattering sources and reduce the accelerated energy of the charge carriers, which prevent the electrical treeing propagation. In this case, the shape of the nanoparticles would affect the enhancement of the electric breakdown strength and spherical nanoparticles are more effective than ellipsoidal or fusiform nanoparticles, so the electric breakdown strength enhancement in ZnO nanocomposites is not significant. From Figure 10, the electric breakdown strength of the nanocomposites all decreases at higher nanoparticle content of 5 phr. This can be explained that with the increase of nanoparticle content, large nanoparticle aggregates may appear and induce void defects around them. Also the distortion of permittivity and conductivity of the nanoparticles and polymer matrix is more obvious when large aggregates appear. As the nanoparticle

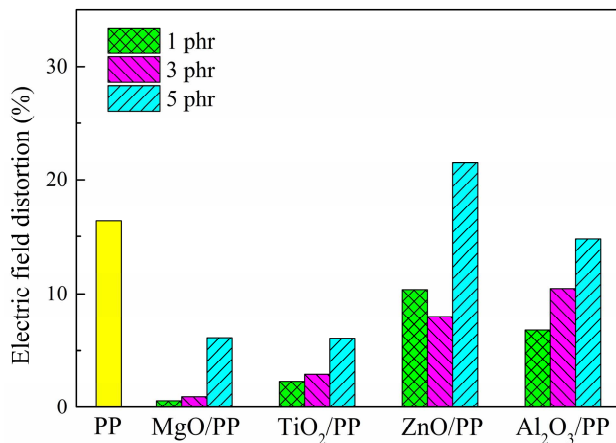


Figure 9. Electric field distortion factor of PP nanocomposites with different nanoparticles after 30 minutes DC electric field poling.

content increases, the inter-particle distance decreases, which help to build up the low-resistance path [31]. From Figure 7 and 8, space charge accumulation and corresponding electric field distortion is more obvious in the nanocomposites with 5 phr nanoparticles, this would also cause the decrease of electric breakdown strength.

3.7 DISCUSSION ON THE FILLER SELECTION RULE

The selection of nanofiller which gives the best results of the nanocomposites has always been considered as the toughest task to the researchers. Now that lots of data have been disclosed on the effect of nanofillers on the properties of the nanocomposites. It is now necessary to establish a filler selection rule to guide the design of polymer nanocomposites. But only the studies of the effect of different nanoparticle types is not sufficient to obtain a clear filler selection rule. The

filler selection rule must be constructed on the comprehensive studies of the physical mechanism behind the increased performance of polymer nanocomposites. But at this stage, the physical mechanism behind is still not very clearly understood. It is reported in literature that properties of the nanocomposites depend on the properties of the nanoparticles (e.g. shape, diameter, permittivity, conductivity and breakdown strength), the properties of the polymer matrix (e.g. permittivity, conductivity, breakdown strength) and the interaction between the nanoparticles and polymer matrix (e.g. the compatibility, dispersion) [4, 32-35]. Also the coupling agent would significantly influence the properties of the nanocomposites [36-38]. This is why it is difficult to establish a filler selection rule to guide the design of polymer nanocomposites with desired properties. We suspect that the matching of the nanoparticles, polymer matrix and coupling agent would play an important role in modifying the properties of the nanocomposites. At this stage, we can only conclude from this study that MgO and TiO₂ nanoparticles is more capable for PP, but clear filler selection rule should be obtained in the future based on the deep understanding of the physics mechanism.

4 CONCLUSIONS

Thermoplastic PP nanocomposites with surface modified MgO, TiO₂, ZnO and Al₂O₃ nanoparticles of various contents were prepared by melt blending. The micro morphology, dielectric properties, DC volume resistivity, space charge behavior and DC electric breakdown strength of the nanocomposites were investigated. The effect of different nanoparticles on the electrical properties of PP was discussed.

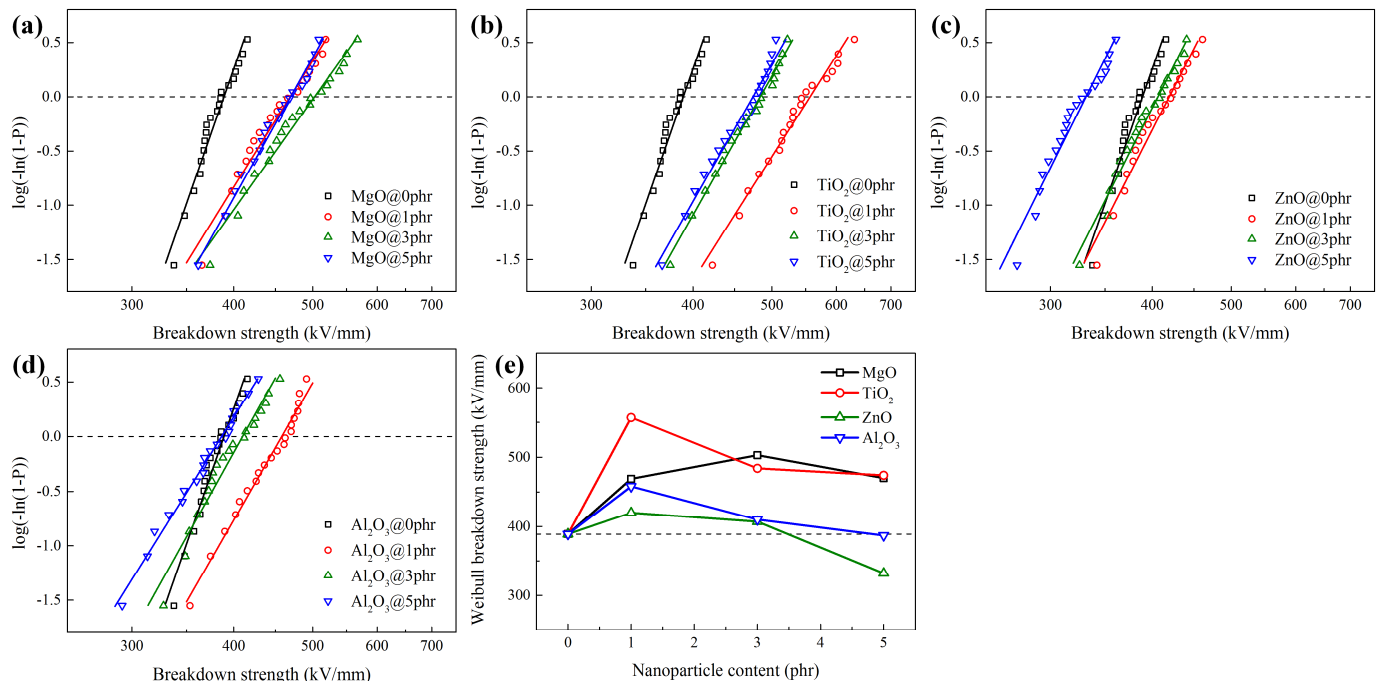


Figure 10. Weibull plots of DC electric breakdown strength of PP nanocomposites with different nanoparticles.

All nanoparticles are well distributed in the PP matrix at lower content as the result of surface modification. With the increase of nanoparticle content, the dielectric permittivity all increases, which should be attributed to the interfacial polarization and the higher permittivity of the nanoparticles. Nanocomposites with MgO, ZnO and Al₂O₃ nanoparticles show slightly lower dielectric loss than pure PP. But the TiO₂/PP nanocomposites exhibit increased dielectric loss.

The DC volume resistivity of MgO, TiO₂ and ZnO nanocomposites all increases with the increase of nanoparticle content firstly and then decreases, while the resistivity of Al₂O₃/PP nanocomposites continues to increase. The increased DC volume resistivity may be attributed to the deep traps at the interfaces between nanoparticles and polymer matrix that reduce the charge carrier mobility and the scattering effects of the nanoparticles. The highest DC volume resistivity of MgO, ZnO and Al₂O₃ nanocomposites is over 3 times of pure PP. But TiO₂ nanoparticles can only slightly increase the DC volume resistivity of PP.

Space charge suppression and DC electric breakdown strength show similar variation trend, both increase with the increase of nanoparticle content and then decrease. Among the four kinds of nanoparticles, MgO and TiO₂ nanoparticles show superior space charge suppression ability. There is almost no space charge in the PP nanocomposites with 3 phr MgO or 1 phr TiO₂ nanoparticles and the DC breakdown strength enhancement for these two nanocomposites is about 43% and 29%, respectively. ZnO nanoparticles have very limited effect on the space charge suppression.

All four kinds of nanoparticles could improve the electrical properties of PP but the degree of enhancement differs. Considering the electrical properties investigated, the optimal content for MgO, TiO₂, ZnO and Al₂O₃ nanoparticles is about 3, 1, 1 and 1 phr, respectively. Among the four kinds of nanoparticles, MgO and TiO₂ nanoparticles show superior ability, which are suggested to be used as nano-fillers to prepare high-performance PP based insulation material for potential recyclable HVDC cables.

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